Prototype Verification Summary — Team 0109D

ESC204 Design Dossier — Artifact 04.4

1 Verification Overview

All prototype subsystems were validated through a structured, three-test empirical protocol followed by a mechanical endurance stress test. The sequence was deliberately ordered by increasing realism and difficulty: **Test 1** (Output Force Quantification) established baseline force magnitudes across nozzle geometries; **Test 2** (Distance & Angle Optimization) characterized clearing performance on a high-friction roof mock-up at varying standoff distances; and **Test 3** (Wet Debris Maximum Stiction) subjected the system to worst-case saturated debris conditions. A final **Endurance Test** verified that the integrated stepper-driven track could sustain continuous automated operation without mechanical or thermal failure.

2 Test 1 — Output Force Quantification

Setup. A Dyson hairdryer at maximum airflow setting directed its output onto a 14×14 cm rigid target resting on a digital scale. Four nozzle geometries were tested at standoff distances of 4 cm, 8 cm, 12 cm, and 16 cm.

Table 1: Measured output force (N) by nozzle geometry and standoff distance.

Nozzle Geometry	**4 cm**	**8 cm**	**12 cm**	**16 cm**
Pure Dyson (Baseline)	1.96	1.95	1.95	1.76
Dyson OEM Air Blade	1.98	1.97	1.95	1.78
Custom Air Blade	1.96	1.95	1.93	1.72
Double-Sided Air Blade	1.91	1.88	1.85	1.62

Key Finding. Total aerodynamic force is conserved at approximately 1.96 N (≈200 g) across all single-exit geometries within 12 cm. Nozzle geometry *redistributes* the momentum flux into a planar shear layer but does not increase it. The sharp degradation observed at 16 cm establishes a hard operational constraint: the nozzle must remain within 12 cm of the target surface.

3 Test 2 — Distance & Angle Optimization

Setup. A 35×15 cm roof mock-up covered in 40-grit aluminum oxide sandpaper (simulating asphalt shingle friction) was locked at a $30°$ incline. A custom alignment frame with protractor ensured nozzle parallelism to the roof pitch; axial alignment was independently verified by a second observer. Disaggregated dry tissue paper served as the debris analogue. Three trials were conducted per configuration at $d = 10$ cm, 25 cm, and 50 cm.

Table 2: Clearing performance: Circular Nozzle vs. Custom Air Blade at $30°$.

Distance	Circular Nozzle	Custom Air Blade
$d = 10$ cm	Full clearance; instantaneous impact force sufficient regardless of geometry.	Full clearance; geometry advantage negligible at close range.
$d = 25$ cm	Partial clearance; central debris ejected but lateral residue remains.	Superior ejection height and distance; planar shear layer penetrates *under* debris.
$d = 50$ cm	Narrow central path cleared only; severe turbulent diffusion leaves lateral residue across the 15 cm width.	Wide, uniform sweep clears full 15 cm width; coherent planar jet resists diffusion.

$30°$ Angle Justification. At $30°$, the airflow vector decomposes into a normal component F_y (sufficient for Coanda boundary-layer attachment without pinning the debris) and a maximized tangential shear component F_x (driving debris down the slope). Angles exceeding $45°$ amplify F_y to the point of pinning; angles below $15°$ cause the stream to deflect over the debris profile without engagement.

4 Test 3 — Wet Debris Maximum Stiction

Setup. Identical to Test 2 at $d = 50$ cm, but with *heavily saturated* tissue paper to maximize static friction and minimize the aerodynamic profile available for dislodgement.

- **Circular Nozzle — FAILED.** The localized high-pressure column pinned wet tissues flat against the 40-grit surface. Only marginal central dislodgement was observed.
  Video: Circular Nozzle Wet Trial
- **Custom Air Blade — PASSED.** The high-velocity planar shear layer slipped beneath the water/sandpaper barrier and peeled saturated debris off the surface.
  Video: Air Blade Wet Trial

Full trial set (8 videos): View Google Drive Folder

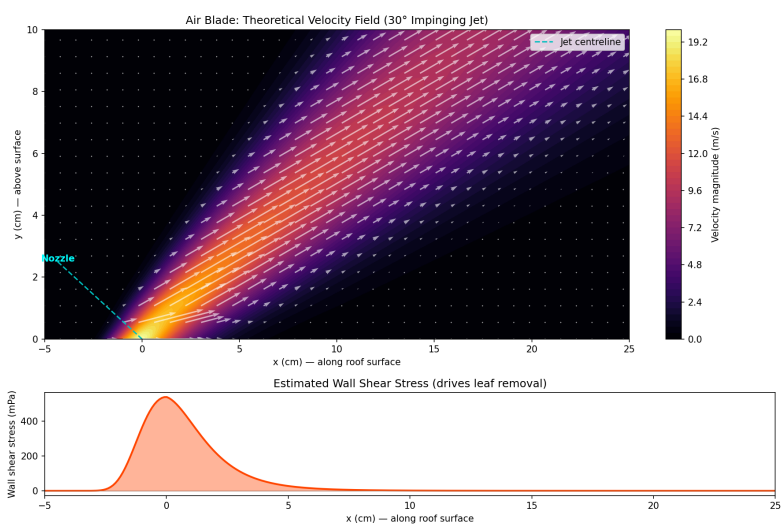

Figure 1: Shear-layer visualization illustrating the planar jet's ability to penetrate under low-profile debris.

5 Endurance Testing

The fully integrated stepper-driven track was subjected to a continuous 2-minute operational cycle, exceeding the minimum 2-pass verification threshold. Results:

- **Stalling:** Zero instances across 120 s of continuous operation.
- **Belt tension:** Maintained through all direction reversals triggered by limit switches.
- **Thermal:** A4988 stepper driver remained within safe operating limits (V_{REF} tuned to ∼50% capacity).
- **Measured traverse:** 56 cm in 22 s $\Rightarrow$ $v \approx 2.55$ cm/s linear carriage velocity.

Video: Endurance Test Folder

6 Specification Compliance

All eleven design specifications were verified and passed. Table 3 summarizes the compliance status.

Table 3: Specification compliance matrix — all requirements satisfied.

Spec ID	Domain	Status
O-1.1-ELEC-1	Electrical	**PASS**
O-2.1-ELEC-1	Electrical	**PASS**
O-2.2-ELEC-1	Electrical	**PASS**
O-3.2-ELEC-1	Electrical	**PASS**
O-1.1-STRU-1	Structural	**PASS**
O-1.1-STRU-2	Structural	**PASS**
O-4.1-STRU-1	Structural	**PASS**
O-4.2-STRU-1	Structural	**PASS**
O-1.3-SOFT-1	Software	**PASS**
O-2.1-SOFT-1	Software	**PASS**
O-3.2-SOFT-1	Software	**PASS**

All supporting video evidence is archived on the team Google Drive and linked inline above.